INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

DIGITAL FORENSIC INVESTIGATION REPORT

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

Case Title: Case Study 3: Investigating Suspected Illegal File Transfer From Memory Evidence	**Course Name:** SBT-DF204 Computer Forensics Case Studies
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Operating Platform: Kali Linux 2026 (Rolling Release)	**Date of Submission:** 18/10/2026
Instructor Name: Aminu Idris	**Report Version:** Version 1.0 (Final Release)

1. Executive Summary

This report documents the forensic investigation of a Windows 7 memory image (memdumpWin7.mem) associated with a suspected illegal file transfer. The investigation acquired and preserved a 1,073,676,288-byte memory image (SHA-256: 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e). Using Volatility 2 (Profile: Win7SP1x86_23418), a timeline of events was reconstructed. The evidence shows that the user IEUser accessed the Downloads folder, plugged in a USB device assigned to the F: drive, and executed the command copy secret_file.docx F:. The console output confirms 1 file(s) copied. The memory capture occurred at 2019-01-06 15:09:06 UTC. The evidence strongly supports the hypothesis of an unauthorized file transfer via removable media, though limitations regarding proof of physical possession are documented.

2. Authority, Scope, and Evidence Description

- **Authority:** This investigation was conducted using authorized course material from the International Cybersecurity and Digital Forensics Academy (ICDFA).
- **Forensic Scope:** Analysis of a Windows 7 physical memory dump to identify evidence of confidential file exfiltration via removable mass storage media.
- **Primary Evidence Artifact:** memdumpWin7.mem (Raw Physical Memory Dump).
- **Evidence Size:** 1,073,676,288 bytes (1,024 MB / 1.0 GB).
- **Cryptographic Hash (SHA-256):** 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e.
- **Working Forensic Duplicate:** memdumpWin7_WorkingCopy.mem (Verified bit-identical hash matching the original evidence).

3. Environment and Acquisition Method

- **Host Operating System:** Kali Linux 2026 (Rolling Release, 64-bit).
- **Memory Forensics Framework:** Volatility Framework 2.6.1 executed under Python 2.7 environment.
- **Cryptographic Utility:** sha256sum for strict dual-copy verification prior to running analysis.
- **Sterile Evidence Ingestion:** The memory dump was acquired via wget into a dedicated evidence/ directory, copied preserving timestamps to working/, and verified before execution.

4. Profile Validation

Executing the Volatility imageinfo plugin against the physical memory capture identified the primary suggested operating system profile as Win7SP1x86_23418. This kernel profile was validated through consistent data parsing across core subsystem plugins (pslist, hivelist, printkey, cmdscan, consoles). The memory capture acquisition timestamp was recorded as 2019-01-06 15:09:06 UTC+0000 (Local Date/Time: 2019-01-06 07:09:06 -0800 PST).

5. Timeline of Events

Chronological event log tracking suspect user activity, hardware device mounting, and file exfiltration in Coordinated Universal Time (UTC):

Time (UTC)	Forensic Event Description	Supporting Volatility Evidence
15:02:54	USB Device Ven_General&Prod_UDisk inserted into system	Registry: USBSTOR LastWrite
15:03:06	USB mass storage device volume assigned to drive F:	Registry: MountedDevices (\DosDevices\F:)
15:04:40	User IEUser navigates to the Downloads directory	ShellBags LastWrite (C:\Users\IEUser\Downloads)
15:06:11	Command Prompt console host (conhost.exe PID 3988) spawned	pslist (cmd.exe PID 3996 / conhost.exe)
~15:06:15	User executes: ipconfig, cd Downloads, copy secret_file.docx F:	cmdscan (CommandHistory buffer)
~15:06:15	Console output registers: '1 file(s) copied.'	consoles (Screen 0xf63b8 dump)
15:09:06	System physical RAM acquired via FTK Imager	imageinfo (Image Date/Time)

6. Answers to Investigation Questions

Definitive answers to the core forensic investigation prompts:

Investigative Question	Forensic Finding / Answer
1. File acquired & preserved?	memdumpWin7.mem (1.0 GB); SHA-256: 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e; working copy verified identical.
2. OS profile and image time?	Profile: Win7SP1x86_23418. Image capture time: 2019-01-06 15:09:06 UTC.
3. User and account artifacts?	Active interactive user: IEUser (SID: S-1-5-21-1716914095-909560446-1177810406-1000, Domain: IE8WIN7).
4. Process and network evidence?	cmd.exe (PID 3996) and conhost.exe (PID 3988). netscan confirmed no active outbound network exfiltration sockets.
5. Command history?	Commands recovered via cmdscan: ipconfig, cd Downloads, copy secret_file.docx F:.
6. Removable-media evidence?	USB mass storage device Ven_General&Prod_UDisk&Rev_5.00 mapped directly to \DosDevices\F:.
7. File Explorer evidence?	ShellBags artifact proves GUI access to C:\Users\IEUser\Downloads

	`at 15:04:40 UTC.`
8. Defensible forensic conclusion?	`HIGH CONFIDENCE: User account IEUser actively copied secret_file.docx from the local Downloads directory to removable USB drive F:.`

7. Challenges and Limitations

- **Account vs. Physical Identity:** Memory forensics conclusively establishes activity under the authenticated user account IEUser; however, in the absence of physical surveillance, it cannot disprove an unlocked workstation scenario.
- **Verification on Destination Media:** While the console buffer confirms 1 file(s) copied, memory forensics alone cannot verify the subsequent physical integrity, retention, or alteration of the file on the external USB storage.
- **Generic USB Device Identifiers:** The connected drive registers as generic 'Ven_General&Prod_UDisk'. Without physical recovery of the flash drive, unique hardware serial tracking is constrained.
- **Timestamp Standardization:** All registry LastWrite timestamps and Volatility internal records are recorded in UTC, requiring careful offset correlation with local host activity.

8. Recommendations

- **Enforce USB Storage Device Whitelisting:** Implement Active Directory Group Policy Objects (GPOs) or Endpoint Detection and Response (EDR) controls restricting USB storage access exclusively to encrypted corporate-issued drives.
- **Enable Advanced Object Access Auditing:** Configure Windows Security Auditing for File System Object Access on sensitive file shares and local user directories to establish persistent event logs (Event ID 4663).
- **Deploy Endpoint Data Loss Prevention (DLP):** Implement DLP software capable of monitoring and blocking file transfer attempts involving designated confidential document classifications (.docx, .pdf, .xlsx).
- **Mandate Screen Lock Policies:** Enforce inactivity timeout screen locks (maximum 5 minutes) paired with physical security training to eliminate unauthorized physical access risks.

9. References

- Volatility Foundation. (2019). Volatility 2.6 Command Reference and Architecture Documentation. Retrieved from https://github.com/volatilityfoundation/volatility
- National Institute of Standards and Technology (NIST). (2021). Guide to Digital Forensics (Special Publication 800-86). U.S. Department of Commerce.
- Ligh, M. H., Case, A., Levy, J., & Walters, A. (2014). The Art of Memory Forensics: Detecting Malware and Threats in Windows, Linux, and Mac Memory. Wiley.
- International Cybersecurity and Digital Forensics Academy (ICDFA). (2026). Computer Forensics Case Studies Courseware.

10. Appendices

Evidence Matrix: Appendix A: Formal Evidence Inventory Log:

ID	Finding Description	Evidence Source	Investigative Significance
E01	Memory Dump Integrity	memdumpWin7.mem	Chain of custody (SHA-256: 500bc1ba...)
E02	Active User Account	Volatile Environment	Proves IEUser was actively logged in
E03	Exfiltration Command	cmdscan plugin	Proves copy command entered in console
E04	Exfiltration Success	consoles plugin	Confirms '1 file(s) copied.' output
E05	Removable Drive Mapping	MountedDevices Key	Proves F: volume mapped to USB device
E06	Folder Navigation Activity	ShellBags Artifacts	Confirms Downloads accessed at 15:04:40 UTC

Command Strings: Appendix B: Key Commands Executed:

```
# Profile Validation & Capture Dissection
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem imageinfo
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 pslist

# User and Account Investigation
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey
-K "Volatile Environment"
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey
-K "Microsoft\Windows NT\CurrentVersion\ProfileList"

# Command History & Screen Buffers
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 cmdscan
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 consoles

# Removable Media & Volume Mapping
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey
-K "ControlSet001\Enum\USBSTOR"
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey
-K "MountedDevices"

# ShellBags Export
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 shellbags
--output=html --output-file=reports/shellbags.html
```

11. Appendix C: Screenshots and Evidence

The following 17 documented figures provide comprehensive forensic visual proof across all phases of the memory investigation:

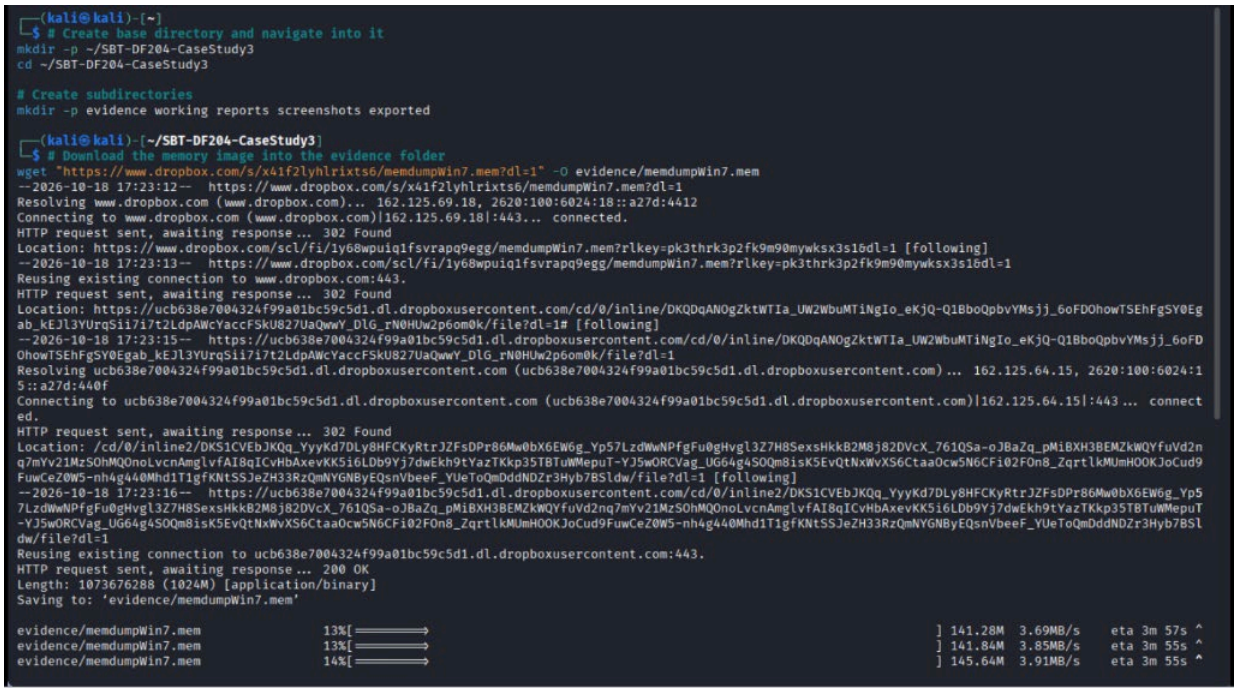


Figure 1: *Evidence Ingestion and Directory Hierarchy Setup* — Initializing /SBT-DF204-CaseStudy3 workspace and acquiring memdumpWin7.mem (1.0 GB) via wget.

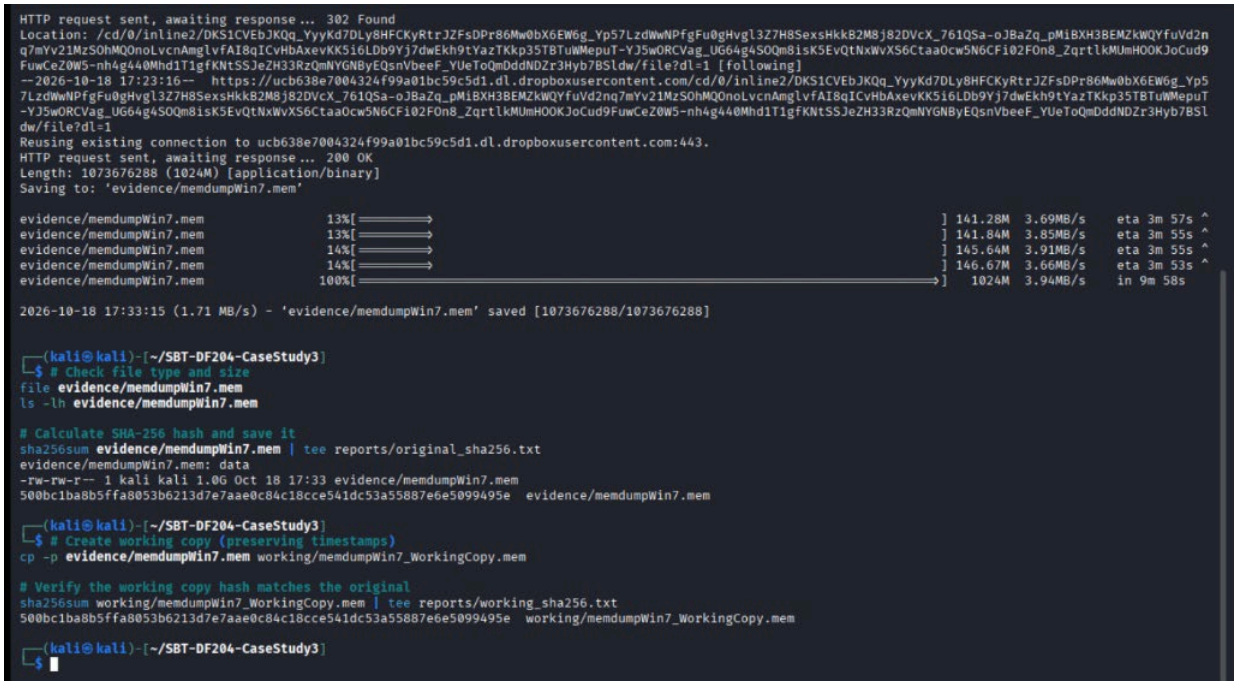


Figure 2: *SHA-256 Integrity Hashing and Working Duplicate Creation* — Computing baseline SHA-256 digest (500bc1ba...), creating working copy, and confirming bit-identical parity.

```
┌──(kali㉿kali)-[~/volatility]
└─$ python2 ~/volatility/vol.py -h
Volatility Foundation Volatility Framework 2.6.1
Usage: Volatility - A memory forensics analysis platform.

Options:
  -h, --help            list all available options and their default values.
                        Default values may be set in the configuration file
                        (/etc/volatilityrc)
  --conf-file=/home/kali/.volatilityrc
                        User based configuration file
  -d, --debug           Debug volatility
  --plugins=PLUGINS     Additional plugin directories to use (colon separated)
  --info                Print information about all registered objects
  --cache-directory=/home/kali/.cache/volatility
                        Directory where cache files are stored
  --cache               Use caching
  --tz=TZ               Sets the (Olson) timezone for displaying timestamps
                        using pytz (if installed) or tzset
  -f FILENAME, --filename=FILENAME
                        Filename to use when opening an image
  --profile=WinXPSP2x86
                        Name of the profile to load (use --info to see a list
                        of supported profiles)
  -l LOCATION, --location=LOCATION
                        A URN location from which to load an address space
  -w, --write           Enable write support
  --dtb=DTB             DTB Address
  --output=text         Output in this format (support is module specific, see
                        the Module Output Options below)
  --output-file=OUTPUT_FILE
                        Write output in this file
  -v, --verbose         Verbose information
  --shift=SHIFT         Mac KASLR shift address
  --physical_shift=PHYSICAL_SHIFT
                        Linux kernel physical shift address
  --virtual_shift=VIRTUAL_SHIFT
                        Linux kernel virtual shift address
  -g KDBG, --kdbg=KDBG  Specify a KDBG virtual address (Note: for 64-bit
                        Windows 8 and above this is the address of
                        KdCopyDataBlock)
  --force               Force utilization of suspect profile
  --cookie=COOKIE       Specify the address of nt!ObHeaderCookie (valid for
```

Figure 3: *Volatility Framework 2.6.1 Environment Verification* — Verifying Volatility 2.6.1 installation, command-line arguments, and plugin availability under Python 2.7.

```
┌──(kali㉿kali)-[~/volatility]
└─$ cd ~/SBT-DF204-CaseStudy3

# Run imageinfo
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem imageinfo | tee reports/imageinfo.txt

# List processes
python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 pslist | tee reports/pslist.txt
Volatility Foundation Volatility Framework 2.6.1
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
                     AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                     AS Layer2 : FileAddressSpace (/home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7_WorkingCopy.mem)
                      PAE type : PAE
                           DTB : 0x185000L
                          KDBG : 0x82972c68L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0x82973d00L
             KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2019-01-06 15:09:06 UTC+0000
     Image local date and time : 2019-01-06 07:09:06 -0800
Volatility Foundation Volatility Framework 2.6.1
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+0000
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+0000
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+0000
0x841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 UTC+0000
0x852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 UTC+0000
0x858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 UTC+0000
0x858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 UTC+0000
0x858bbd28 lsass.exe               440    340      7      581      0      0 2019-01-06 18:02:45 UTC+0000
0x858be2c0 lsm.exe                 448    340     10      147      0      0 2019-01-06 18:02:45 UTC+0000
0x858f88d8 svchost.exe             552    432     10      359      0      0 2019-01-06 18:02:45 UTC+0000
0x8590cb00 VBoxService.ex          616    432     13      123      0      0 2019-01-06 18:02:45 UTC+0000
0x85913030 svchost.exe             672    432      8      260      0      0 2019-01-06 15:02:47 UTC+0000
0x85761030 svchost.exe             724    432     18      420      0      0 2019-01-06 15:02:47 UTC+0000
0x858d9208 svchost.exe             828    432     21      484      0      0 2019-01-06 15:02:47 UTC+0000
0x859d6518 svchost.exe             880    432     31      983      0      0 2019-01-06 15:02:47 UTC+0000
0x84f992c0 svchost.exe             992    432     13      275      0      0 2019-01-06 15:02:48 UTC+0000
0x85a23ce8 svchost.exe            1064    432     14      367      0      0 2019-01-06 15:02:48 UTC+0000
0x85a4fd28 spoolsv.exe            1184    432     12      272      0      0 2019-01-06 15:02:48 UTC+0000
```

Figure 4: *OS Profile Identification (imageinfo) & Process Dissection* — imageinfo plugin determining profile Win7SP1x86_23418 and capture timestamp 2019-01-06 15:09:06 UTC.

```
0x859d6518 svchost.exe             880    432     31      983      0      0 2019-01-06 15:02:47 UTC+0000
0x84f992c0 svchost.exe             992    432     13      275      0      0 2019-01-06 15:02:48 UTC+0000
0x85a23ce8 svchost.exe            1064    432     14      367      0      0 2019-01-06 15:02:48 UTC+0000
0x85a4fd28 spoolsv.exe            1184    432     12      272      0      0 2019-01-06 15:02:48 UTC+0000
0x85a91030 svchost.exe            1228    432     17      315      0      0 2019-01-06 15:02:48 UTC+0000
0x85ade488 vmicsvc.exe            1364    432      4       94      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae3030 vmicsvc.exe            1388    432      5      105      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae9030 taskhost.exe           1412    432      8      145      1      0 2019-01-06 15:02:48 UTC+0000
0x85af6d28 vmicsvc.exe            1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+0000
0x85b06d28 vmicsvc.exe            1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+0000
0x850553f0 vmicsvc.exe            1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+0000
0x85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0x859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0x859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
0x859bab00 taskeng.exe            1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
0x8500bd00 cygrunsrv.exe          1984   1680      0 ------      0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0x84202030 conhost.exe            2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0x84203998 sshd.exe               2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
0x8501cce8 GoogleUpdate.e          192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+0000
0x85b11398 sppsvc.exe             1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+0000
0x85a07030 svchost.exe            2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+0000
0x857ce6d0 rundll32.exe           2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+0000
0x84fddd28 dwm.exe                2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+0000
0x85bba860 explorer.exe           2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+0000
0x85c7f030 VBoxTray.exe           2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+0000
0x85cb9820 SearchIndexer.         2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+0000
0x85e287f0 WUDFHost.exe           3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+0000
0x84fa2ad0 msiexec.exe            3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+0000
0x841eb658 WmiPrvSE.exe           4032    552      5      114      0      0 2019-01-06 15:03:54 UTC+0000
0x8583a030 chrome.exe             2388   2524     33      819      1      0 2019-01-06 15:04:30 UTC+0000
0x851e8520 chrome.exe             1280   2388      7       82      1      0 2019-01-06 15:04:30 UTC+0000
0x850316b8 chrome.exe             2380   2388      2       57      1      0 2019-01-06 15:04:30 UTC+0000
0x852845c8 svchost.exe            2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+0000
0x842b4d28 chrome.exe             2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+0000
0x858be710 svchost.exe            3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+0000
0x8431fb08 cmd.exe                3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+0000
0x842ddd28 conhost.exe            3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+0000
0x842eb6b0 chrome.exe              976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+0000
0x84312030 FTK Imager.exe         2596   2524     14      358      1      0 2019-01-06 15:07:15 UTC+0000
0x850d5a38 dllhost.exe            2228    552      6 170 ... 38      1      0 2019-01-06 15:09:07 UTC+0000

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

Figure 5: *Active Process Tree Listing (pslist)* — Inspecting active processes: cmd.exe (PID 3996), conhost.exe (PID 3988), and FTK Imager.exe (PID 2596).

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 hivelist | tee reports/hivelist.txt
Volatility Foundation Volatility Framework 2.6.1
Virtual    Physical   Name
---------- ---------- ----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "Volatile Environment" | tee reports/volatile
_environment.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: Volatile Environment (V)
Last updated: 2019-01-06 15:02:48 UTC+0000

Subkeys:
  (V) 1

Values:
REG_SZ        LOGONSERVER     : (V) \\IE8WIN7
REG_SZ        USERDOMAIN      : (V) IE8WIN7
REG_SZ        USERNAME        : (V) IEUser
REG_SZ        USERPROFILE     : (V) C:\Users\IEUser
REG_SZ        HOMEPATH        : (V) \Users\IEUser
REG_SZ        HOMEDRIVE       : (V) C:
```

Figure 6: *Registry Hive Listing and Volatile Environment Extraction* — hivelist plugin locating ntuser.dat and printkey extracting active user IEUser (Domain: IE8WIN7).

```
Values:
REG_SZ        LOGONSERVER     : (V) \\IE8WIN7
REG_SZ        USERDOMAIN      : (V) IE8WIN7
REG_SZ        USERNAME        : (V) IEUser
REG_SZ        USERPROFILE     : (V) C:\Users\IEUser
REG_SZ        HOMEPATH        : (V) \Users\IEUser
REG_SZ        HOMEDRIVE       : (V) C:
REG_SZ        APPDATA         : (V) C:\Users\IEUser\AppData\Roaming
REG_SZ        LOCALAPPDATA    : (V) C:\Users\IEUser\AppData\Local

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "Microsoft\Windows NT\CurrentVersion\ProfileL
ist" | tee reports/profilelist.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:
  (S) S-1-5-18
  (S) S-1-5-19
  (S) S-1-5-20
  (S) S-1-5-21-1716914095-909560446-1177810406-1000
  (S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:
REG_EXPAND_SZ ProfilesDirectory : (S) %SystemDrive%\Users
REG_EXPAND_SZ Default           : (S) %SystemDrive%\Users\Default
REG_EXPAND_SZ Public            : (S) %SystemDrive%\Users\Public
REG_EXPAND_SZ ProgramData       : (S) %SystemDrive%\ProgramData

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "Software\Microsoft\Windows\CurrentVersion\Ex
plorer\MountPoints2" | tee reports/mountpoints2.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
```

Figure 7: *User ProfileList and MountPoints2 Inspection* — Inspecting user SIDs under ProfileList and examining user MountPoints2 registry keys.

```
----------
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:
  (S) S-1-5-18
  (S) S-1-5-19
  (S) S-1-5-20
  (S) S-1-5-21-1716914095-909560446-1177810406-1000
  (S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:
REG_EXPAND_SZ ProfilesDirectory : (S) %SystemDrive%\Users
REG_EXPAND_SZ Default           : (S) %SystemDrive%\Users\Default
REG_EXPAND_SZ Public            : (S) %SystemDrive%\Users\Public
REG_EXPAND_SZ ProgramData       : (S) %SystemDrive%\ProgramData

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2" | tee reports/mountpoints2.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
  (S) CPC
  (S) F
  (S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
  (S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
  (S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
  (S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
  (S) {a5b8a983-608c-11e5-a266-806e6f6e6963}

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

Figure 8: *User MountPoints2 Drive F: Association* — Examining MountPoints2 subkeys confirming drive letter F: mapped with LastWrite 2019-01-06 15:03:07 UTC.

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 cmdscan | tee reports/cmdscan.txt
Volatility Foundation Volatility Framework 2.6.1
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2???2
Cmd #37 @ 0x3200c4: 3?4?2???2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 consoles | tee reports/consoles.txt
Volatility Foundation Volatility Framework 2.6.1
**************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
```

Figure 9: *Command History Buffer Extraction (cmdscan)* — cmdscan recovering command execution sequence: ipconfig, cd Downloads, and copy secret_file.docx F:.

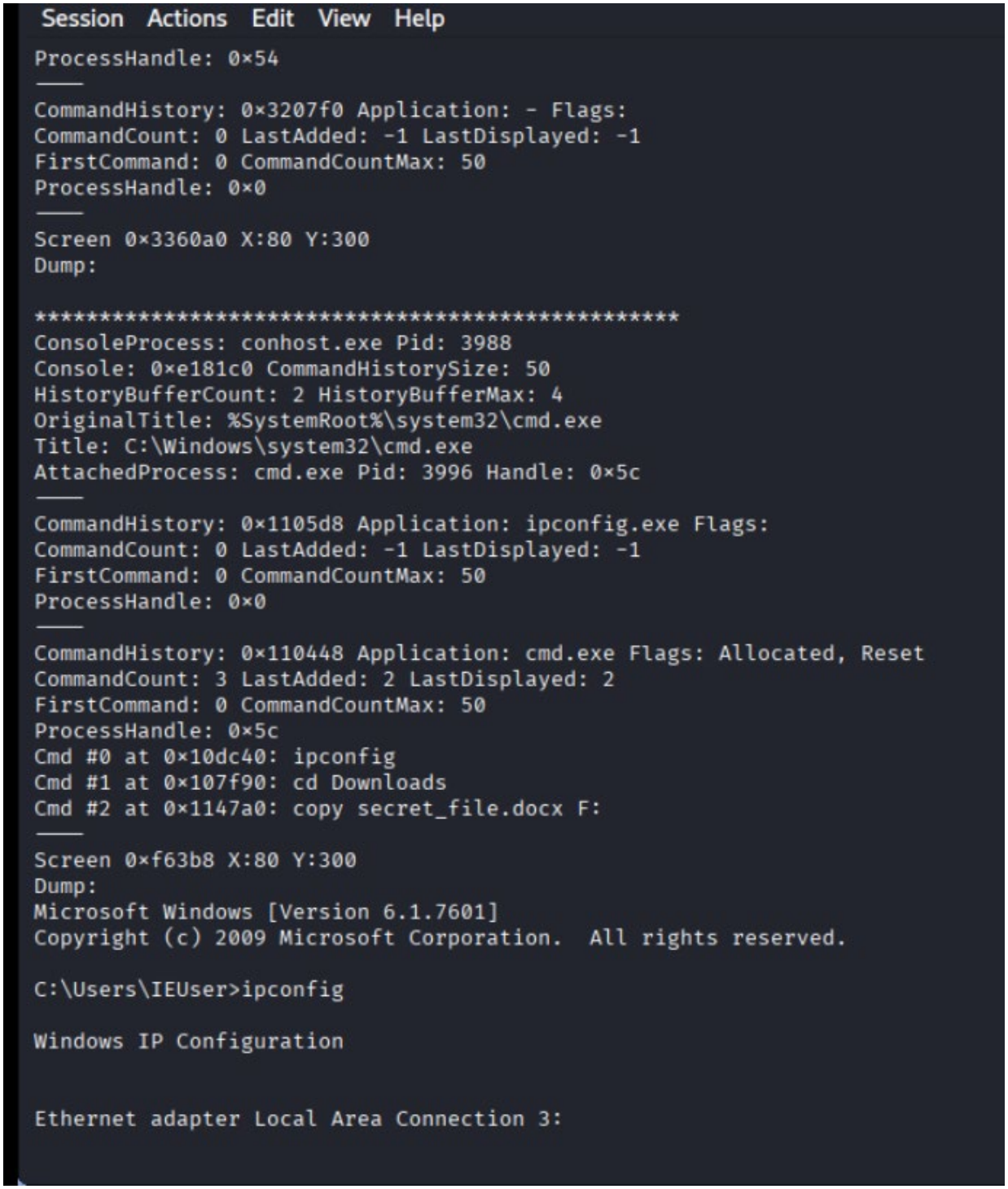


Figure 10: *Console Process Reassembly (consoles Start)* — consoles plugin reconstructing cmd.exe (PID 3996) screen buffer and initial IP configuration output.

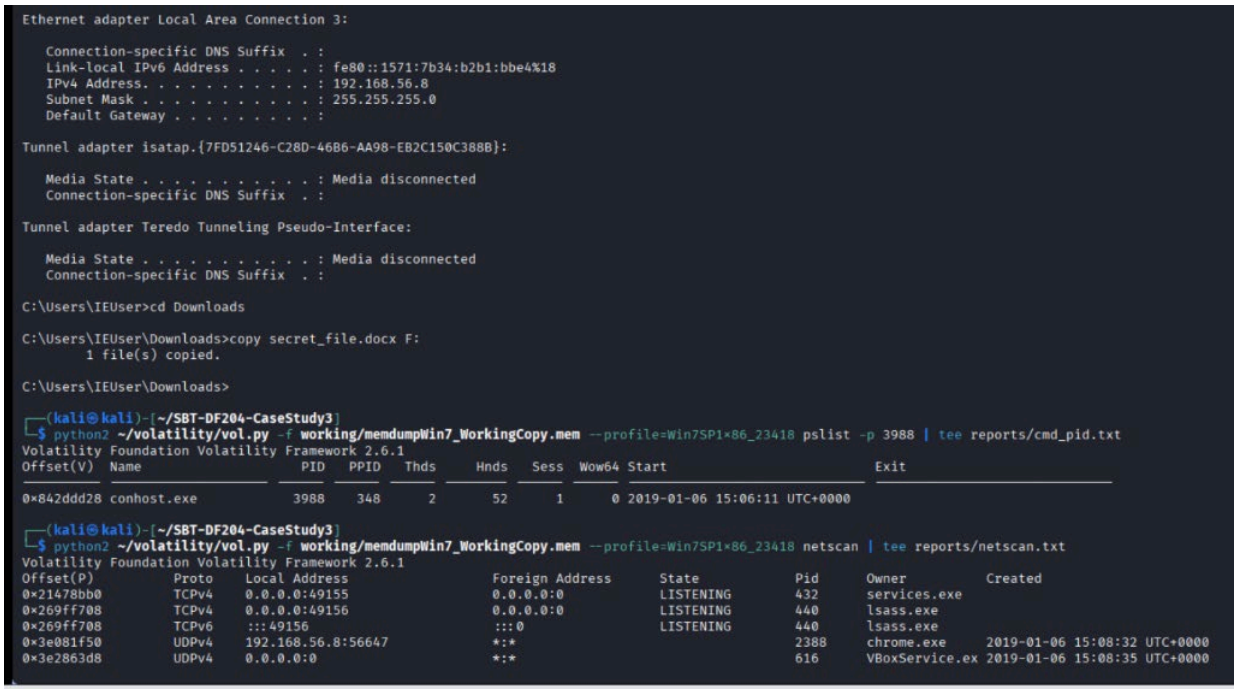


Figure 11: *Definitive File Copy Confirmation ('1 file(s) copied.')* — Console screen buffer showing execution of copy secret_file.docx F: and confirmation output: 1 file(s) copied.

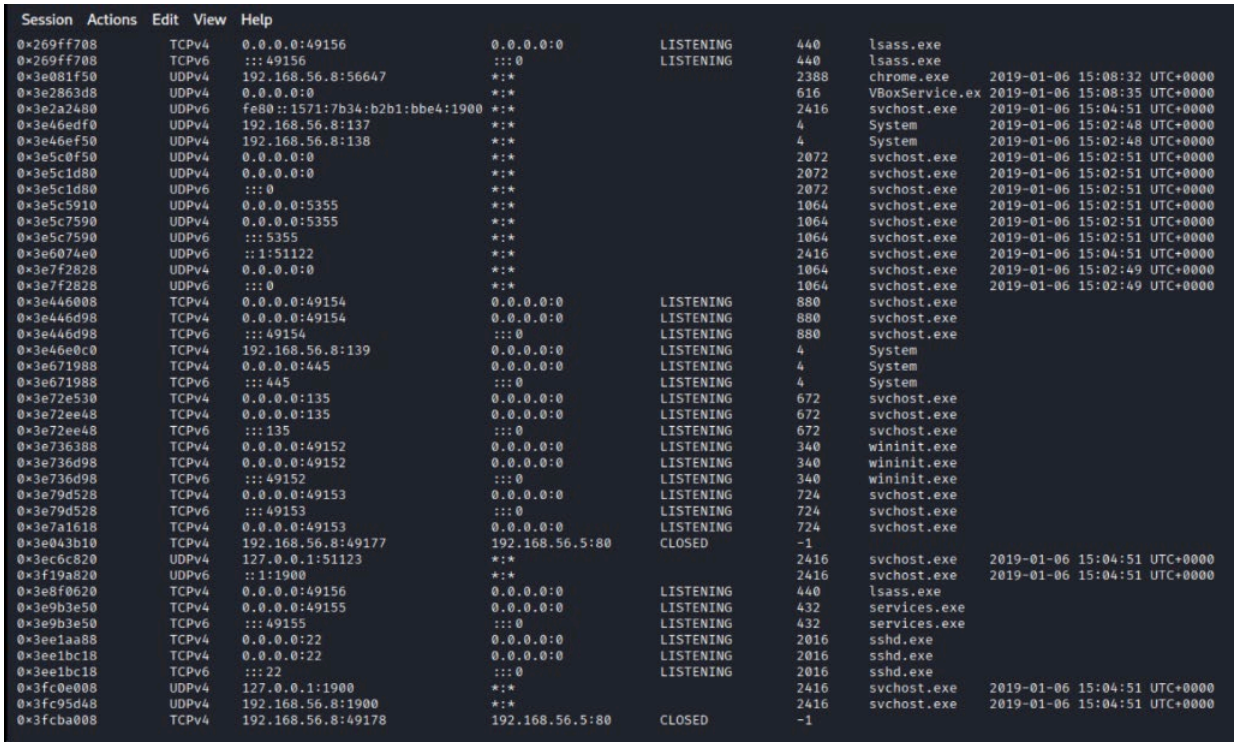


Figure 12: *Network Sockets and Listener Analysis (netscan)* — netscan plugin confirming no active external sockets associated with cmd.exe or conhost.exe.

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Enum\USBSTOR" | tee reports/us
bstor.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) Disk&Ven_General&Prod_UDisk&Rev_5.00

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "ControlSet001\Control\DeviceClasses\{53f5630
7-b6bf-11d0-94f2-00a0c91efb8b}" | tee reports/deviceclasses.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: {53f56307-b6bf-11d0-94f2-00a0c91efb8b} (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) ##?#IDE#DiskVBOX_HARDDISK_____________________________1.0_____#56106af1716061.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
  (S) ##?#IDE#DiskVirtual_HD________________________________1.1.0___#5635dc70406060.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
  (S) ##?#IDE#DiskVMware_Virtual_IDE_Hard_Drive_____________00000001#562eba496060.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
  (S) ##?#SCSI#Disk&Ven_Dell&Prod_VIRTUAL_DISK#6&17b13437&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
  (S) ##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_60#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "MountedDevices" | tee reports/mounteddevices
.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile
```

Figure 13: *Removable Media Identification (USBSTOR Registry)* — printkey USBSTOR confirming insertion of Ven_General&Prod_UDisk&Rev_5.00 at 2019-01-06 15:02:54 UTC.

```
└─$ python2 ~/volatility/vol.py -f working/memdumpWin7_WorkingCopy.mem --profile=Win7SP1x86_23418 printkey -K "MountedDevices" | tee reports/mounteddevices
.txt
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: MountedDevices (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

Values:
REG_BINARY    \DosDevices\C:  : (S)
0x00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                ...P........
REG_BINARY    \??\Volume{a5b8a980-608c-11e5-a266-806e6f6e6963} : (S)
0x00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                ...P........
REG_BINARY    \DosDevices\D:  : (S)

REG_BINARY    \DosDevices\E:  : (S)
0x00000000  5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00   \.?.?.\.I.D.E.#.
0x00000010  43 00 64 00 52 00 6f 00 6d 00 56 00 42 00 4f 00   C.d.R.o.m.V.B.O.
0x00000020  58 00 5f 00 43 00 44 00 2d 00 52 00 4f 00 4d 00   X._.C.D.-.R.O.M.
0x00000030  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000040  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000050  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000060  5f 00 5f 00 5f 00 5f 00 5f 00 31 00 2e 00 30 00   _._._._._._1...0.
0x00000070  5f 00 5f 00 5f 00 5f 00 5f 00 23 00 35 00 26 00   _._._._._._#.5.&.
0x00000080  33 00 39 00 34 00 63 00 30 00 61 00 64 00 33 00   3.9.4.c.0.a.d.3.
0x00000090  26 00 30 00 26 00 30 00 2e 00 31 00 2e 00 30 00   &.0.&.0...1...0.
0x000000a0  23 00 7b 00 35 00 33 00 66 00 35 00 36 00 33 00   #.{.5.3.f.5.6.3.
0x000000b0  30 00 64 00 2d 00 62 00 36 00 62 00 66 00 2d 00   0.d.-.b.6.b.f.-.
0x000000c0  31 00 31 00 64 00 30 00 2d 00 39 00 34 00 66 00   1.1.d.0.-.9.4.f.
0x000000d0  32 00 2d 00 30 00 30 00 61 00 30 00 63 00 39 00   2.-.0.0.a.0.c.9.
0x000000e0  31 00 65 00 66 00 62 00 38 00 62 00 7d 00         1.e.f.b.8.b.}.
REG_BINARY    \??\Volume{a5b8a985-608c-11e5-a266-806e6f6e6963} : (S)
0x00000000  5c 00 3f 00 3f 00 5c 00 46 00 44 00 43 00 23 00   \.?.?.\.F.D.C.#.
0x00000010  47 00 45 00 4e 00 45 00 52 00 49 00 43 00 5f 00   G.E.N.E.R.I.C._.
0x00000020  46 00 4c 00 4f 00 50 00 50 00 59 00 5f 00 44 00   F.L.O.P.P.Y._.D.
0x00000030  52 00 49 00 56 00 45 00 23 00 35 00 26 00 33 00   R.I.V.E.#.5.&.3.
0x00000040  61 00 61 00 33 00 39 00 34 00 37 00 65 00 26 00   a.a.3.9.4.7.e.&.
0x00000050  30 00 26 00 30 00 23 00 7b 00 35 00 33 00 66 00   0.&.0.#.{.5.3.f.
0x00000060  35 00 36 00 33 00 30 00 64 00 2d 00 62 00 36 00   5.6.3.0.d.-.b.6.
```

Figure 14: *MountedDevices Registry Key Extraction (System Volumes)* — printkey MountedDevices displaying drive mappings for C:, D:, and internal system volumes.



Figure 15: *MountedDevices Volume to USBSTOR Correlation* — MountedDevices binary mapping Volume{421e7e52...} directly to USBSTOR General UDisk hardware.

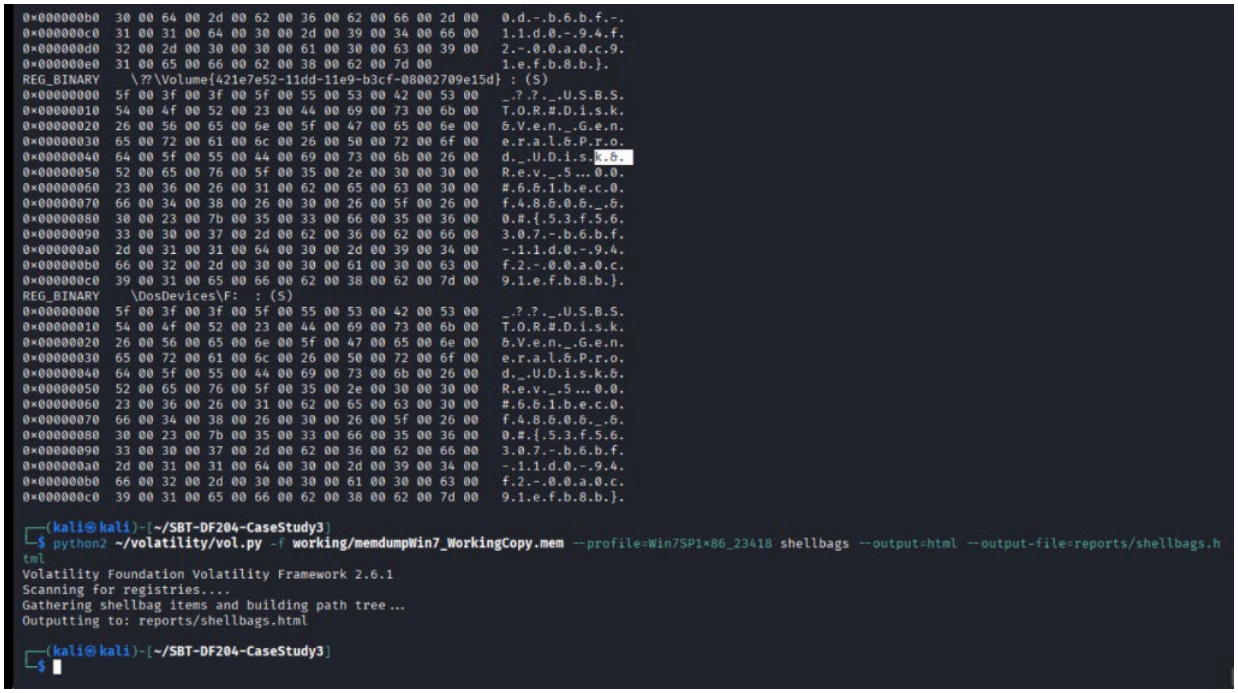


Figure 16: *Drive Letter Assignment (\DosDevices\F:)* — Confirming \DosDevices\F: maps directly to USBSTOR General UDisk device and initiating ShellBags export.

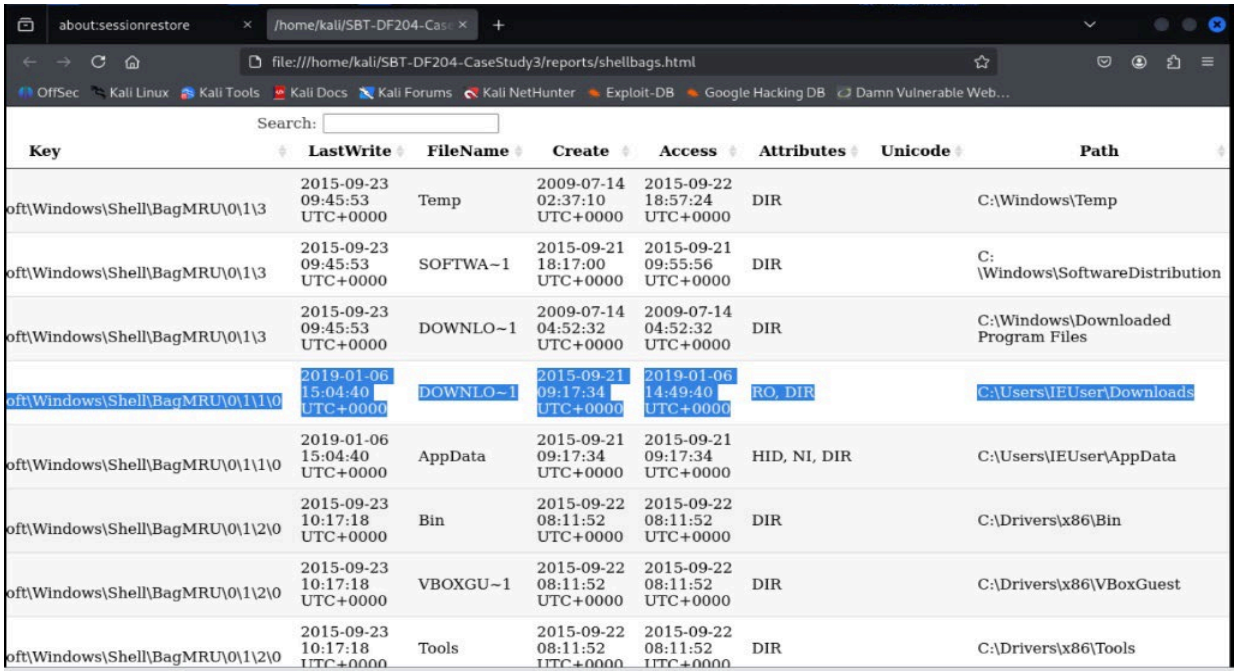


Figure 17: *ShellBags GUI HTML Report Dissection* — HTML ShellBags export confirming user IEUser navigated to C:\Users\IEUser\Downloads at 15:04:40 UTC.

Student Declaration

> I confirm that I completed this investigation in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.
>
> Student Signature: Almustapha Yusuf Date: 18/10/2026